An Overview of Computer Animation

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This Section's Objectives

- Explore what computer animation is about
- Provide an historical introduction

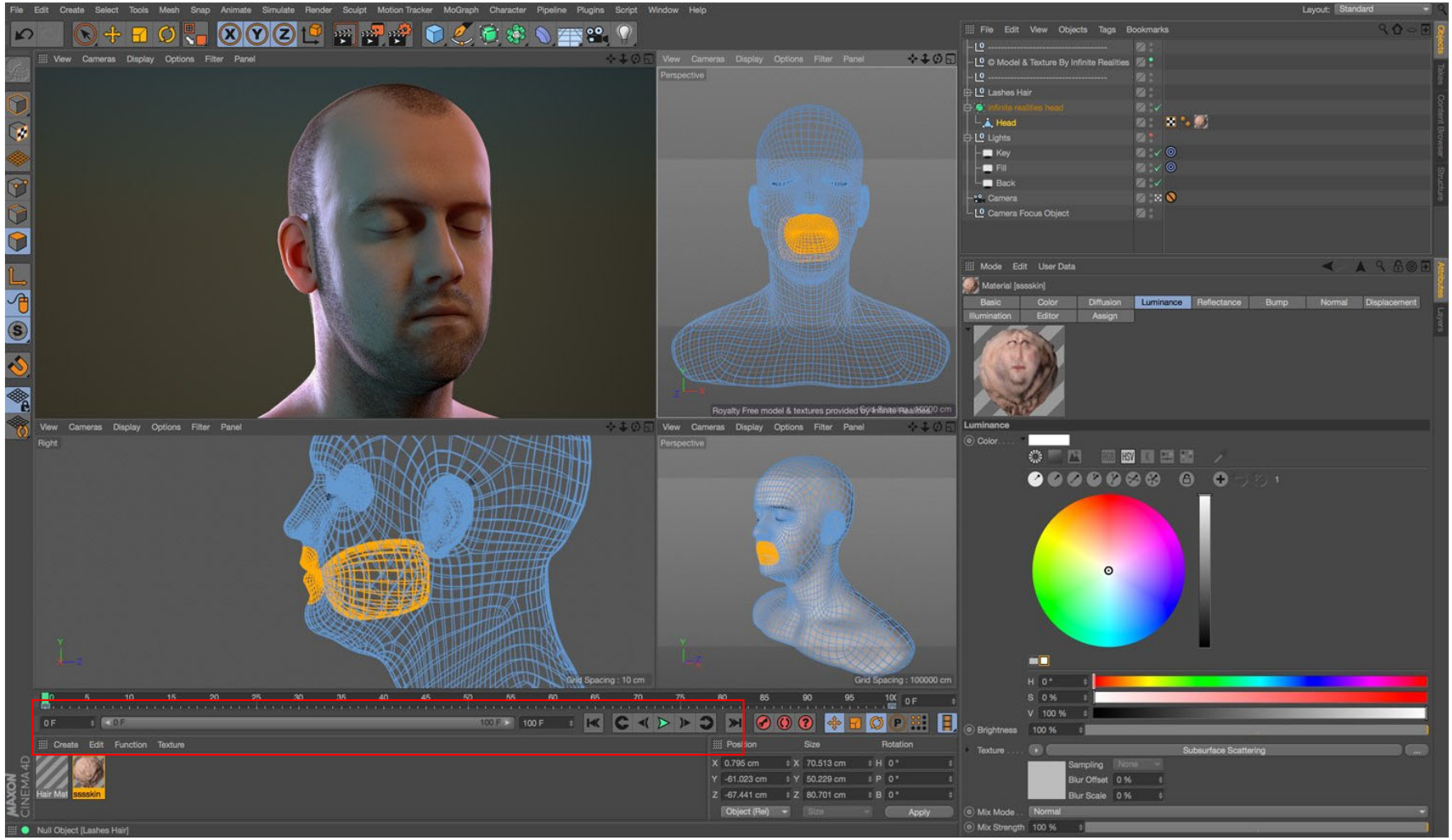
Types of Computer Animation

- Three Categories:
  - Keyframe Animation
  - Procedural Animation
  - Motion Captured Animation

Types of Computer Animation

- Keyframe Animation
  - Most of the work is done by the animator
  - s/he explicitly creates the animation by setting keyframe positions of the models over time
  - the software interpolates the keyframes
  - requires lots of skill

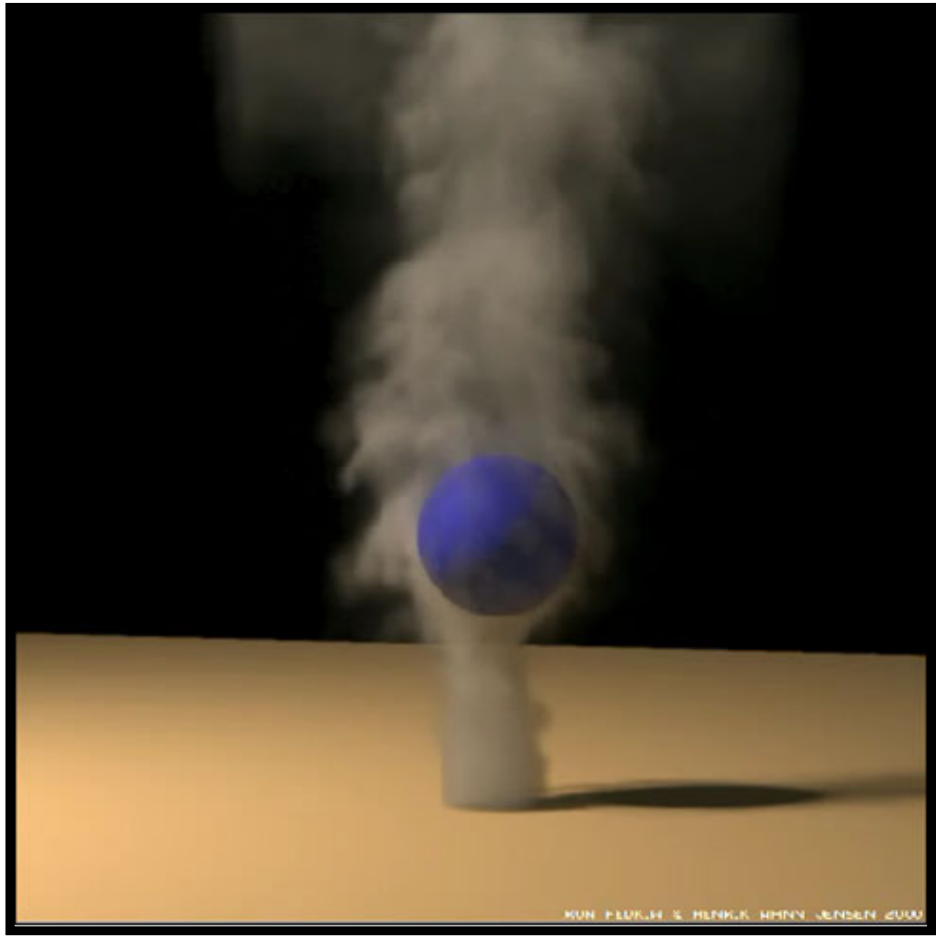
Keyframe Animation



Types of Computer Animation

- Procedural Animation
  - Most of the work is done by the computer
  - Objects positions and shapes are controlled by:
    - Particle, Flocking and Crowd Systems
    - Cloth, Fire, Smoke and Water simulations

Procedural Animation



Fedkiw & Wann Jensen [video]

Types of Computer Animation

- Motion Captured Animation
  - Most of the work is done by a human actor
  - Human joint angles are taken from a live model via sensors on a motion capture suit
  - Often needs touching up:
    - Filtering, editing, retargeting, stitching,…

Motion Capture

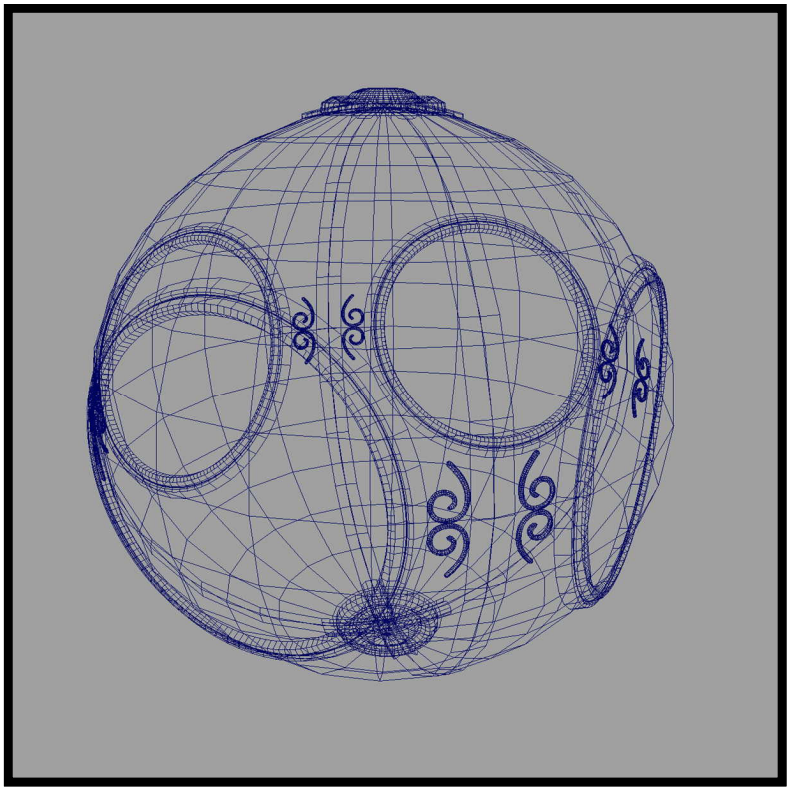
Pyros Pictures [video]

History of CA – 1960's

- Early pioneering work
- Wireframe graphics
- Project Sketchpad

History of CA – 1960's

- Wireframe graphics



History of CA – 1960's

Project Sketchpad

- Ivan Sutherland's PhD thesis at MIT
- Interactive display loop
  - Display something
  - User moves light pen
  - Computer regenerates display

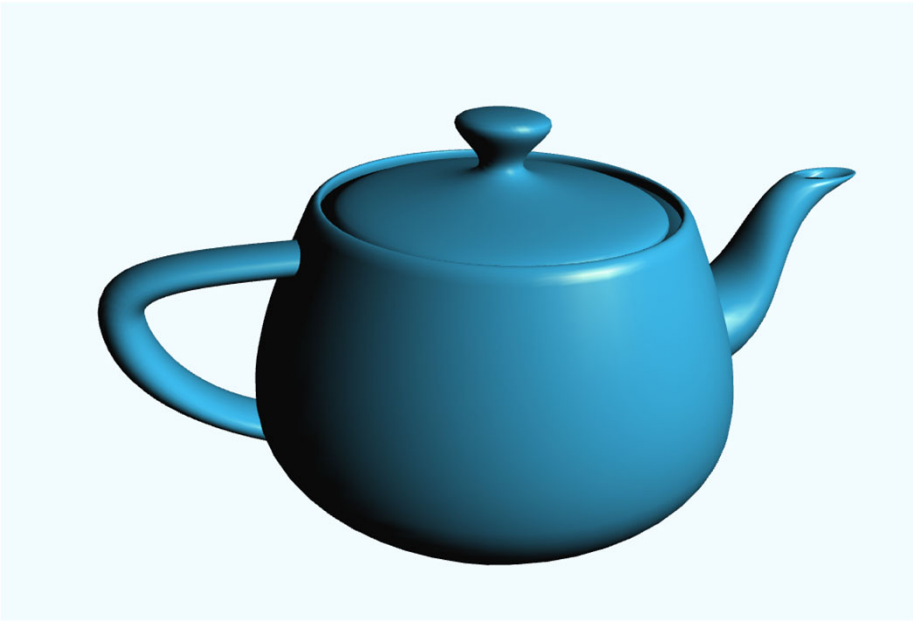
History of CA – 1970's

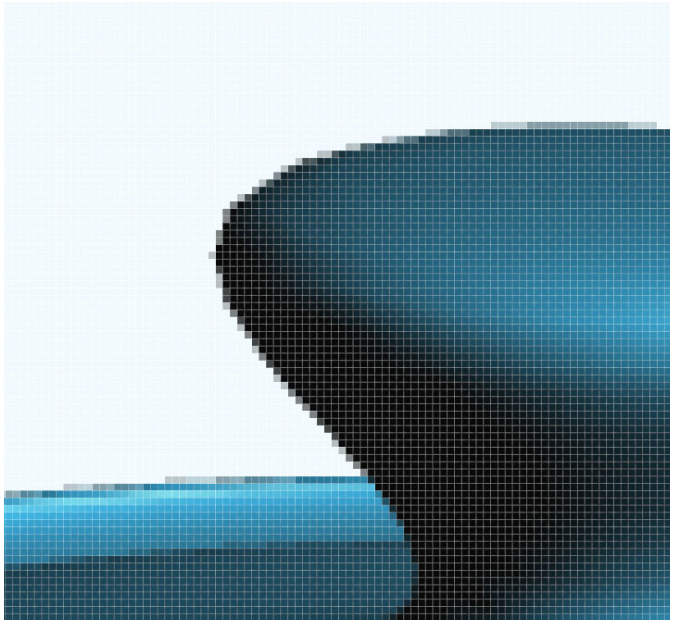
- Introduction of raster graphics
- Beginning of graphics standards
  - GKS: European effort
    - Becomes ISO 2D standard
  - Core: North American effort
    - 3D but fails to become ISO standard
- Graphics moves onto workstations and PCs

History of CA – 1970's

Raster Graphics

- Image produced as an array of picture elements (*pixels*) in the *frame buffer* – not just lines

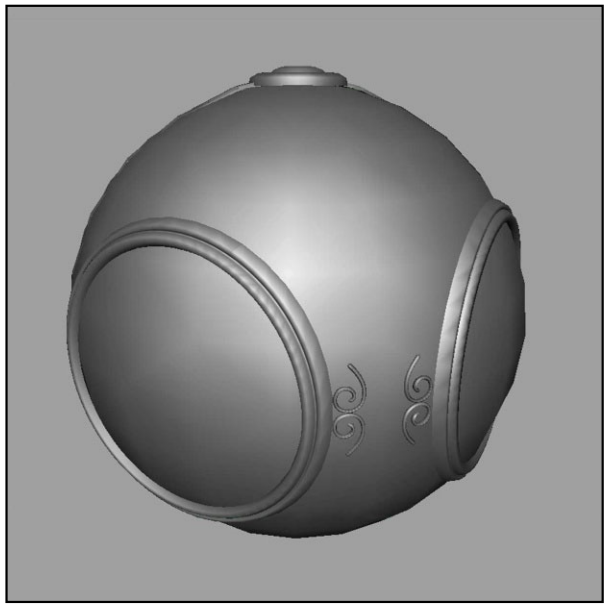




History of CA – 1970's

New rendering algorithms

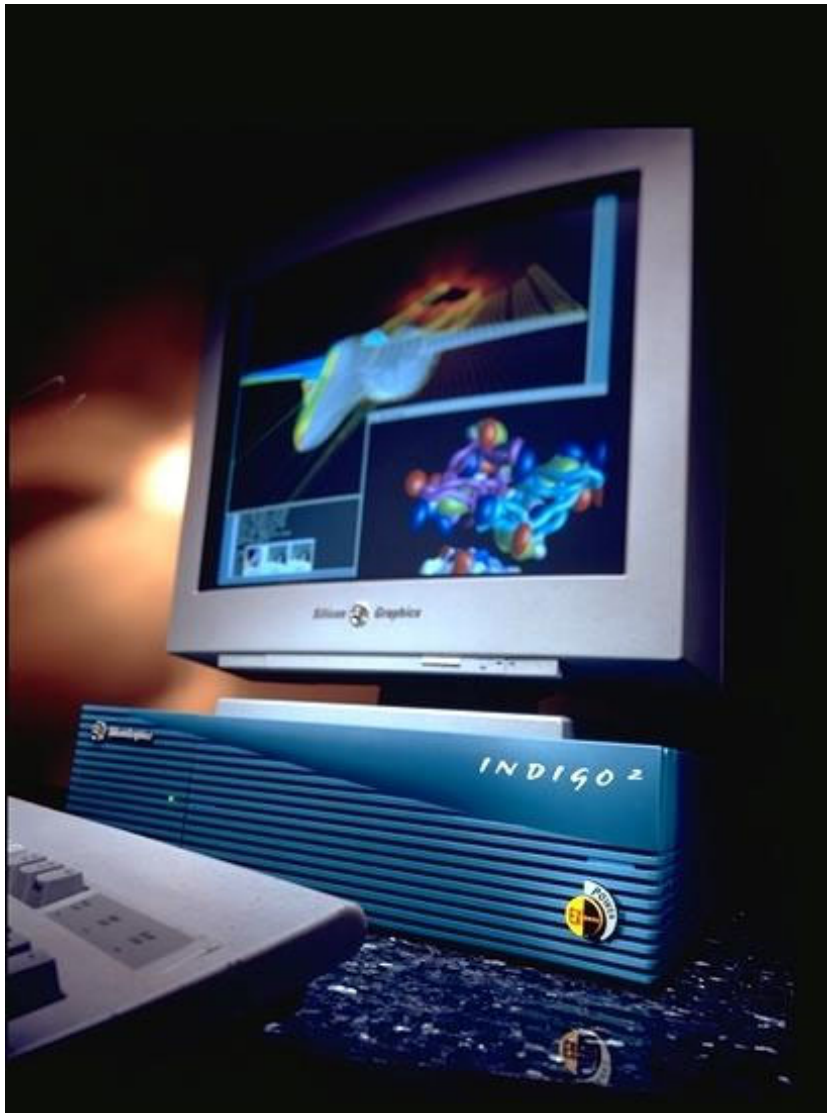
- Gouraud shading
- Phong shading
- Curved surfaces
- Anti-aliasing



smooth shading

History of CA – 1980's

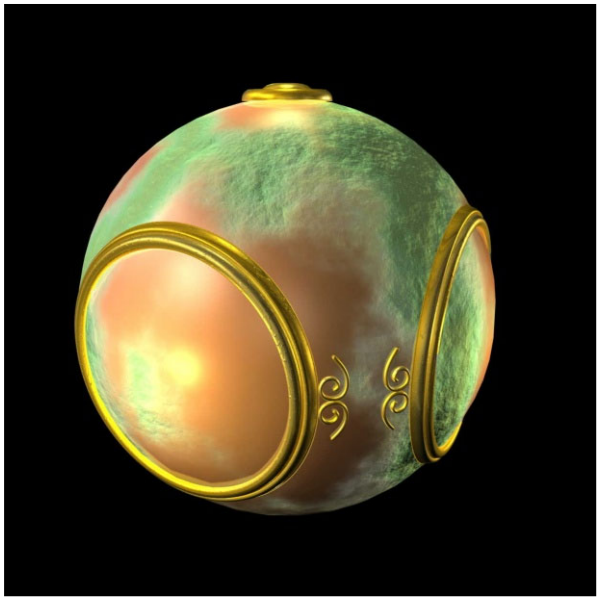
- Special purpose hardware
  - Silicon Graphics 3D geometry engine
- Industry-based standards
  - PHIGS
  - RenderMan
- Texture, environment and bump mapping
- Global illumination



History of CA – 1980's

Finally, some realism(?) comes to computer graphics!

environmental mapping

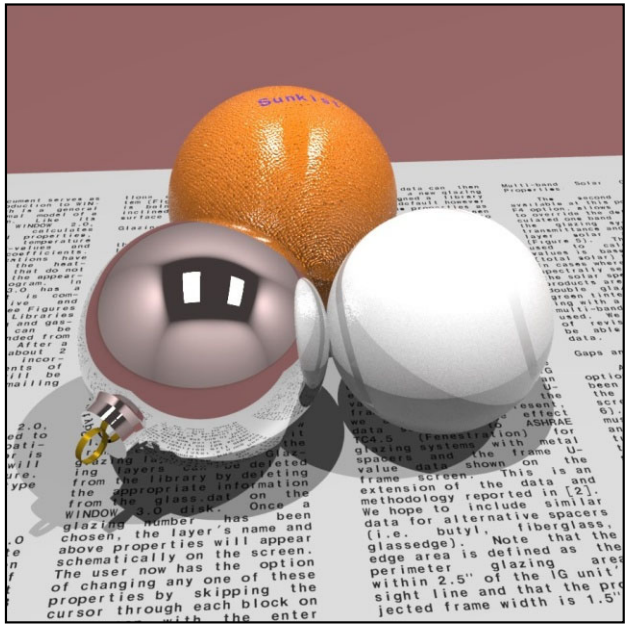


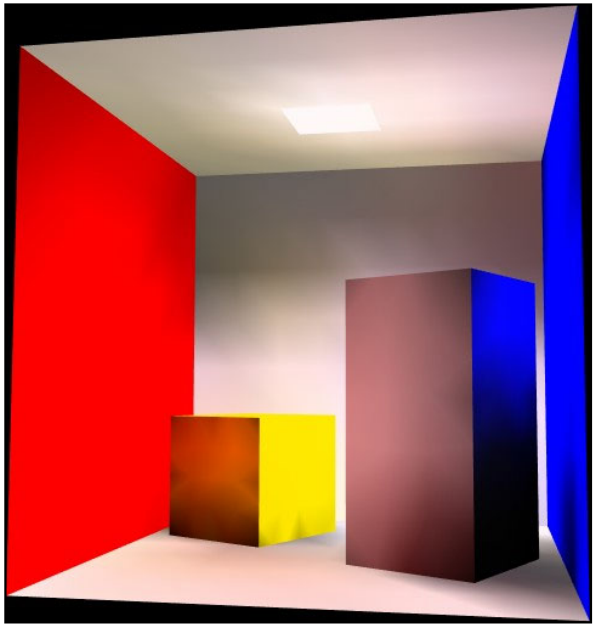
bump mapping

History of CA – 1980's

Global illumination

- Ray tracing
- Radiosity





History of CA – 1990's

- OpenGL API
- Completely computer-generated feature-length movies (Toy Story) are successful
- New techniques:
  - Non-photorealistic images
  - QuickTime VR
  - Cloth, fire animation
  - Human animation

History of CA – 1990's

Non-photorealistic imagery

History of CA – 1990's

Cloth animation

Choi & Ko [video]

History of CA – 1990's

QuickTime VR

History of CA – 2000+

- Graphics cards for PCs dominate market
- Gaming systems determine direction of market
- Computer graphics routine in movie industry
- Blurring of Hardware and Software

- New techniques
  - More Non-photorealistic Processing and Animation
  - Sub-surface scattering
  - Motion capture databases

Blurring of H/W and S/W

Programmable Hardware

- Graphics Processing Units (GPUs)
- More of the rendering pipeline is placed onto the GPU

Vertex Shaders

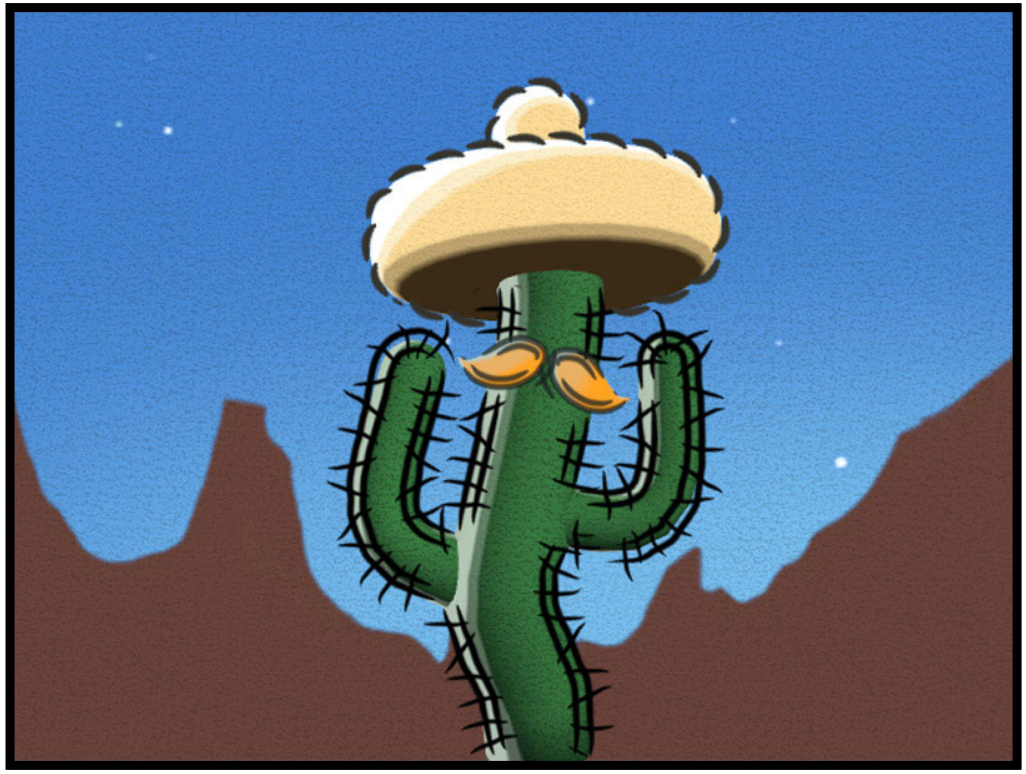
- are graphics processing functions which manipulate object positions on the GPU

Pixel Shaders

- are small programs which process pixels on the GPU

History of CA – 2000+

Non-photorealistic animation



Kalnins, Markosian, Meier, Kowalski, Lee, Davidson, Webb, Hughes, Finkelstein [video]

History of CA – 2000+

Recent Stuff!

Title Slide Images from Canadian Researchers

Image references, clockwise from top left:

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